

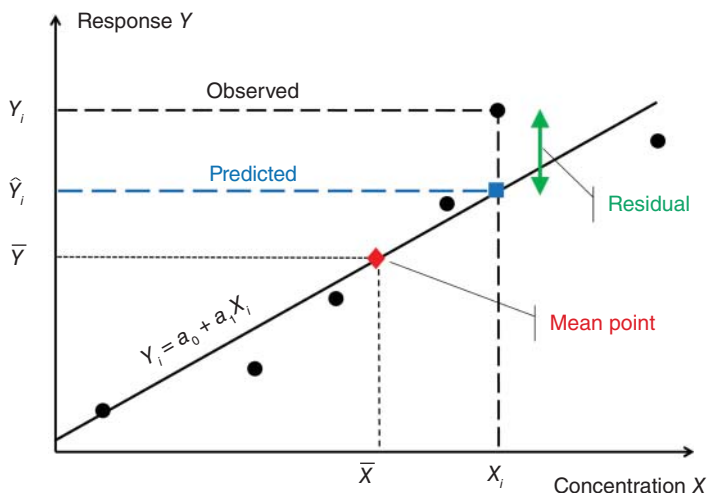


Figure 2.3 Principles of ordinary least-squares (OLS) method.

But, to apply the OLS method, it must be assumed that E_X is negligible compared to E_Y and explicitly set to zero ($E_X = 0$). In other words, it is assumed that calibrators are prepared with no error. As explained in Section 6.6.2, it is manageable to estimate the MU for each calibrator and verify if it is null or negligible. Given the way calibrators are usually prepared, this assumption seems reasonable as a first approximation. The major downside is that it is always possible to apply OLS whenever the basic assumption is wrong, but the interpretation of the obtained coefficients may be erroneous.

The principle of the OLS method is to estimate the coefficients a_0 and a_1 by minimizing the sum of the squared differences between observed and predicted values, called residuals, or also deviates, and expressed by Eq. (2.5). Such a difference can be geometrically interpreted as a distance symbolized by a double green arrow in Figure 2.3. The first idea could be to calculate the average of residuals and try to make it as small as possible. Intuitively, it is easy to understand that any line shall pass by the mean point reported as a red diamond on the graphics.

Unfortunately, if the selected line regularly passes through all points, the simple average distance shall always be zero because the sum of negative residuals shall be exactly balanced by that of positive residuals. To avoid this inconvenience, the residuals are generally squared. This numerical solution gave its name to the method: least-squares regression. Other solutions exist, such as summing the absolute differences, but are not as common as the least-squares algorithm. It can be summarized by the following equations:

Residual

$$E_i = Y_i - \hat{Y}_i \quad (2.5)$$

Condition on the sum of squared residuals

$$S = \sum_{i=1}^I E_i^2 \rightarrow \text{Minimum} \quad (2.8)$$

In the case of a straight-line, Eqs. (2.6) and (2.7) are combined to obtain the residual, and Eq. (2.8) is modified as follows:

The sum of squared residuals (for the straight line)

$$S = \sum_{i=1}^I (Y_i - a_0 - a_1 X_i)^2 \rightarrow \text{Minimum} \quad (2.9)$$

Equation (2.9) is a function with two unknowns a_0 and a_1 since all values X_i and Y_i are known. A function passes through minimum at the point where the first derivative is zero. The first two derivatives, one with respect to a_0 and the other with respect to a_1 are calculated and set to 0. A set of simultaneous equations with two unknowns is obtained and solved. After simplification, the calculation comes down to the following set of formulas directly applicable while all elements are known. Notation SS for sum of squares and SP for sum of crossed products are used for simplicity.

Mean point

$$\bar{X} = \frac{\sum_{i=1}^I X_i}{I} \quad \bar{Y} = \frac{\sum_{i=1}^I Y_i}{I} \quad (2.10)$$

Sum of crossed products of deviates

$$SP(X, Y) = \sum_{i=1}^I (X_i - \bar{X})(Y_i - \bar{Y}) \quad (2.11)$$

Sum of squared deviates for X

$$SS(X) = \sum_{i=1}^I (X_i - \bar{X})^2 \quad (2.12)$$

Slope or sensitivity

$$a_1 = \frac{SP(X, Y)}{SS(X)} \quad (2.13)$$

Intercept or blank

$$a_0 = \bar{Y} - a_1 \bar{X} \quad (2.14)$$

Variance of the residuals or residual variance

$$\left\{ \begin{aligned} s_E^2 &= \frac{\sum_{i=1}^I (Y_i - \hat{Y}_i)^2}{I - 2} \\ s_E^2 &= \frac{SS(Y) - a_1 SP(X, Y)}{I - 2} \end{aligned} \right. \quad (2.15)$$

Residual standard deviation

$$s_E = \sqrt{\frac{SS(Y) - a_1 SP(X, Y)}{I - 2}} \quad (2.16)$$

2.2.2 Assumptions and Complements

These results are established under the following set of assumptions:

- Variables E_i are randomly distributed according to Normal laws $\mathcal{N}(0, \sigma^2)$ with zero means and theoretical variances σ^2 . All laws associated with E_i are identical throughout the calibration domain. This means all response variances for all calibrators (or standard solutions) are constant and equal to σ^2 . The best estimate of σ^2 is equal to s_E^2 (Eq. 2.15). If this assumption is not acceptable, a modified method called weighed least-squares (WLS) must be used, as explained in Section 2.3.2.
- Variables E_i are independent, which means that they are not correlated. To comply with this assumption, it is recommended to separately prepare the calibrators. The common incorrect practice of making successive dilutions can exacerbate this dependence and must be discouraged. Despite this well-known tip, many commercial kit manuals still advise making successive dilutions.
- Outliers must be eliminated beforehand. The OLS method, far from pointing out model errors or nonconforming values, will force the model to pass through all points. Moreover, the use of a sum of squares as a minimization criterion gives much greater weight, or leverage, to any point which is far from the others and can thus very easily introduce a bias; it is not a robust method.

When these assumptions are considered acceptable (although rarely verified), it is then possible to calculate various additional parameters described by Eqs. (2.17)–(2.22). The residual standard deviation s_E allows to obtain other standard deviations, such as the blank, the sensitivity, or a predicted value $\hat{Y}$ and the associated confidence intervals. Ultimately, s_E gives a global indication of the fitting closeness between the predicted line and the experimental points.

Standard deviation of sensitivity

$$s_{a_1} = \frac{s_E}{\sqrt{SS(X)}} \quad (2.17)$$

Standard deviation of analytical blank

$$s_{a_0} = s_E \sqrt{\left(\frac{1}{I} + \frac{\bar{X}^2}{SS(X)} \right)} \quad (2.18)$$

Confidence interval of sensitivity

$$\left[a_1 \pm t_{1-\frac{\alpha}{2}; I-2} \times s_{a_1} \right] \quad (2.19)$$

Confidence interval of blank

$$\left[a_0 \pm t_{1-\frac{\alpha}{2}; I-2} \times s_{a_0} \right] \quad (2.20)$$

Quantile of Student's t for a confidence level of $1 - \alpha$, $I - 2$ degrees of freedom

$$t_{1-\frac{\alpha}{2}; I-2}$$

Predicted response for a given X value

$$\hat{Y} = a_0 + a_1 X \quad (2.21)$$

Standard deviation of a predicted response

$$s_{\hat{Y}} = s_E \sqrt{\left(\frac{I+1}{I} + \frac{(X_k - \bar{X})^2}{SS(X)} \right)} \quad (2.22)$$

The confidence intervals as given by Eqs. (2.19) and (2.20) are applied in Section 2.4.3 to present a simplified method for verifying the parallelism of two straight lines. Confidence intervals and tests of hypothesis are procedures for making so-called statistical inferences, i.e. the domain of statistics devoted to making global decisions, or inferring a conclusion, from limited information obtained on a sample. It is considered that hypothesis tests are seldom needed if confidence intervals are available because they are similarly effective and can be graphically interpreted. When two confidence intervals are overlapping, this means that the statistical parameters are not significantly different, or when a confidence interval contains a given value, such as 0, the related parameter is not significantly different from this value.

Up to now, least-squares algorithm is described for the straight-line case, i.e. the model containing one first-order explicative variable denoted X . Multiple regression models can also be handled by OLS and similar formulas obtained. To demonstrate how the OLS algorithm works in the case of more complicated multivariate models, it requires using mathematical matrix notation and calculus. An example is presented in Section 1.5.1, even if this is out of the scope of this book. Full description is available in many statistical textbooks, e.g. the excellent reference book of Draper and Smith about regression [2].

In the next chapter, an example of the quadratic model is presented, and it shows how the coefficients can easily be computed with Excel. Once the coefficients are estimated, the inverse calibration function can be used to obtain the inverse-predicted concentration of unknown samples. Let us denote Y_k the instrumental response recorded with an unknown sample k and Z_k its inverse-predicted concentration given by Eq. (2.24) in the case of a straight-line:

Inverse – predicted concentration (general model)

$$Z = f^{-1}(Y) \quad (2.23)$$

Inverse – predicted concentration (linear model)

$$Z_k = \frac{Y_k - a_0}{a_1} \quad (2.24)$$

As explained in Section 1.1.1, to simplify notation but emphasize the difference between X_c the concentration in the calibrator is controlled by the experimenter and $\hat{X}$ the concentration of the authentic analyte predicted for an unknown sample, the inverse-predicted concentration is denoted Z in the following chapters. However, a more straightforward model for Z could be $Z = g(Y)$. This model defines what can be called *reverse* calibration. It would mean that calibration is no longer a two-step procedure. The inverse calibration function becomes useless to predict the concentration of unknown samples, whereas reverse prediction model could directly accomplish this.

The pros and cons of the reverse model are explained in Section 2.5. Depending on the quantification procedure, the principal benefit of this notation is to emphasize the idea that it is not always obvious that $g = f^{-1}$. It must also be noted that the standard deviation of a predicted response s_{Y_k} given by Eq. (2.22) must not be confounded with the standard deviation s_Z of the inverse-predicted concentration. The basic reason is that s_{Y_k} is related to Y and expressed in the same units as the instrumental response, while s_Z is related to the concentration X and in the analyte unit. The possible estimation of this important standard deviation is addressed in Section 2.5.1 as well as the diverse problems raised.

2.3 Software Implementation

2.3.1 Ordinary Least-squares (OLS) Regression

With any worksheet, the coefficients of the model described by the Eq. (2.6) can be estimated by using the built-in function `LINEST`. An example of application to the data of Table 2.1 is presented in the worksheet named Resource A.

Some explanation will help in understanding this worksheet. Beforehand, calibration data of Table 2.1 must be unfolded into two columns and stored in cells A5 : B14. In cell A16, the built-in function `=LINEST (B5 : B14 ; A5 : A14 ; TRUE ; TRUE)` is typed. Results returned by `LINEST` are illustrated below in the case of a simple straight-line regression with two coefficients. To make the worksheet easily recycled by any analyst, formulas are visible in the column on the right of the cell, giving the result. For instance, the formula applied in cell A22 is visible in cell C22. The interpretation of the results appearing in range A16 : B20 is more complex, as explained below.

Following Excel nomenclature, `LINEST` is a matrix (or tabular) function, i.e. it requires one or more matrices as arguments and returns several results as a matrix (in the mathematical meaning of the word). To exactly understand the way arguments are entered and results returned, it is necessary to look at the user's manual.

	a1	a0	
Coefficients	12.634	0.580	
s_{a_1}	0.332	1.530	s_{a_0}
r^2	0.995	3.981	s_E
F -value	1452.238	8	Degrees of freedom
Regression SS	23011.141	126.762	Residual SS

According to the Excel version, the procedure to apply matrix functions could differ. In recent releases, typing the function in one cell allows us to obtain the results automatically distributed in a set of cells. If there is no place for this operation, an error is emitted. In older versions, it was necessary to first select the destination range and simultaneously type three keystrokes Shift, Ctrl, and Enter. Whatever the